Universität Siegen

MECHATRONICS SYSTEM LABORATARY

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MANIPULATOR ROBOT-COLOR SORTING WITH A MANIPULATOR AND A CONVEYOR

GROUP 6

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1. Abstract

This project presents the analysis and implementation of a mechatronic system for automated color sorting using an existing manipulator robot integrated with a conveyor system. The system combines mechanical components, sensors, actuators, control algorithms, and software to perform coordinated pick-and-place operations. The manipulator consists of rigid links driven by electric motors through gearboxes and is equipped with encoders and touch sensors for joint homing and position feedback. A conveyor system with a color sensor is utilized to detect and transport colored objects to predefined stations.

Inverse kinematics are derived using a geometric approach for accurate positioning. The system is programmed in EV3 Micro Python with sensor-based control and PID concepts to perform color-based sorting and pick-and-place operations, demonstrating effective mechatronic system integration.

2. Introduction

In this project, a Lego EV3 Mindstorms manipulator robot is used to perform automated pick-and-place and color sorting operations. Pick-and-place robots are a class of industrial manipulators widely employed in automation systems for material handling, sorting, and assembly. These robots improve efficiency, accuracy, and repeatability in processes where objects must be identified and placed at predefined locations.

2.1 Classification of Pick-and-Place Robots

Pick-and-place robots can be broadly classified based on their functionality as:

- **Task-Specific**
  Robots designed to perform a predefined operation in a specific environment.
- **Multi-Purpose**
  Reprogrammable robots capable of performing multiple tasks through software modification.

The LEGO EV3 robot used in this project belongs to the **multi-purpose robot category**.

2.2 Types of Pick-and-Place Robots

- **Gantry Robots**: Operate along linear axes and are suitable for heavy-load applications.
- **Articulated Robots**: Use rotary joints, providing high flexibility and precision.
- **SCARA Robots**: Designed for high-speed horizontal assembly operations.
- **Delta Robots**: Parallel robots optimized for high-speed lightweight pick-and-place tasks.

Based on its structure and motion characteristics, the LEGO EV3 robot is best classified as a serial articulated manipulator robot.

2.3 Description of the Robot

Actuators

- The manipulator is driven by three electric motors with encoders, and the conveyor system is driven by one motor with an encoder.

Sensors

- The system uses two touch sensors to perform joint homing and enforce motion limits, three encoders to measure the angular positions of the joints, and a single-color sensor to detect the color of balls moving along the conveyor.

Mechanical Structure

- The manipulator consists of five links: Link-0 serves as the base link, while Link-1, Link-2, Link-3, and Link-4 have lengths of 50 mm, 95 mm, 185 mm, and 110 mm respectively. In terms of kinematic constraints, Link-1 is rigidly connected to Link-2 at an angle of 135°, Link-4 remains perpendicular to the ground plane, and the rotation of Link-3 is mechanically limited by Link-2 on the lower side and by a touch sensor on the upper side. Additionally, the base rotation is constrained using a touch sensor.

Operating Area

- The workspace consists of five stations: Station 1 represents the base reference position, Station 2 is the opposite base position, Station 3 corresponds to the left end of the conveyor, Station 4 to the right end of the conveyor, and Station 5 to the middle of the conveyor.

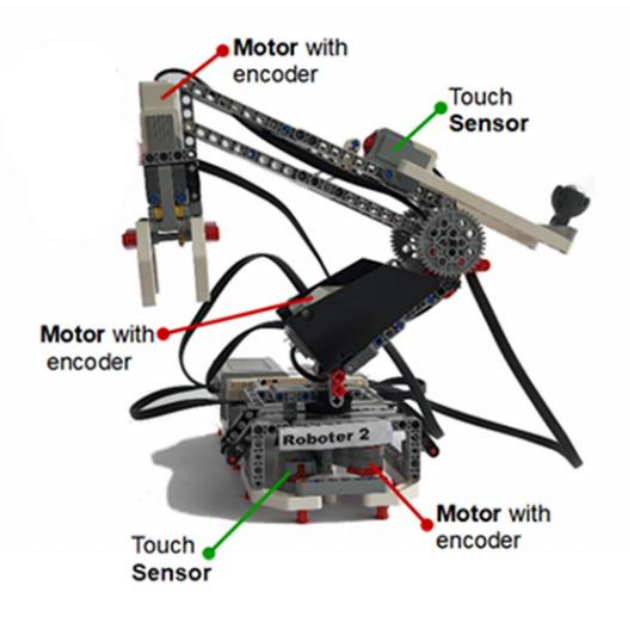


Fig 1

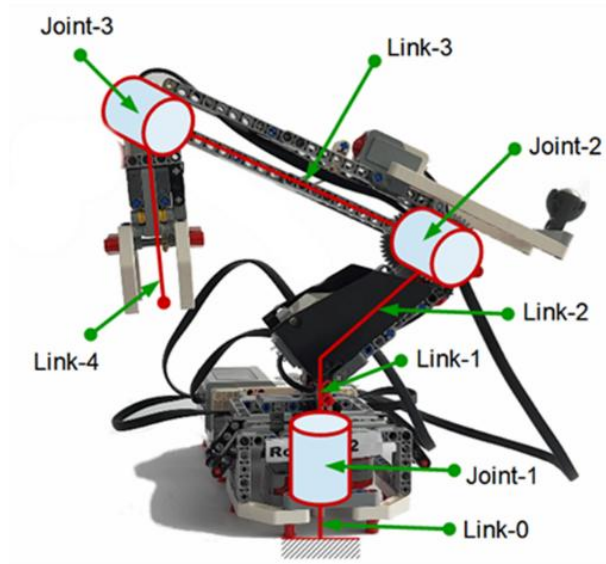


Fig 2

3. Kinematic Modeling of the Manipulator

- Used to calculate the joint angles required to reach a desired end-effector position.
- A geometric approach is applied based on known link lengths and kinematic constraints.
- Trigonometric relationships are used to compute joint angles for predefined target stations.
- Enables accurate positioning of the manipulator during pick-and-place operations.

3.1 Calculation of Link 0

Link-0 was measured using a ruler as the distance from the ground plane to the center of the first joint axis. The measured length of Link-0 is **55 mm**.

3.2 Equation of θ_1

θ_1 represents Link 1's rotational movement around the Z-axis. Figure 1 illustrates the physical significance of the manipulator when viewed from above.

We can obtain $\boldsymbol{\theta_1}$ by using tan $\boldsymbol{\theta}$:

$\theta_1 = \tan^{-1}(y/x)$ ---- (1)

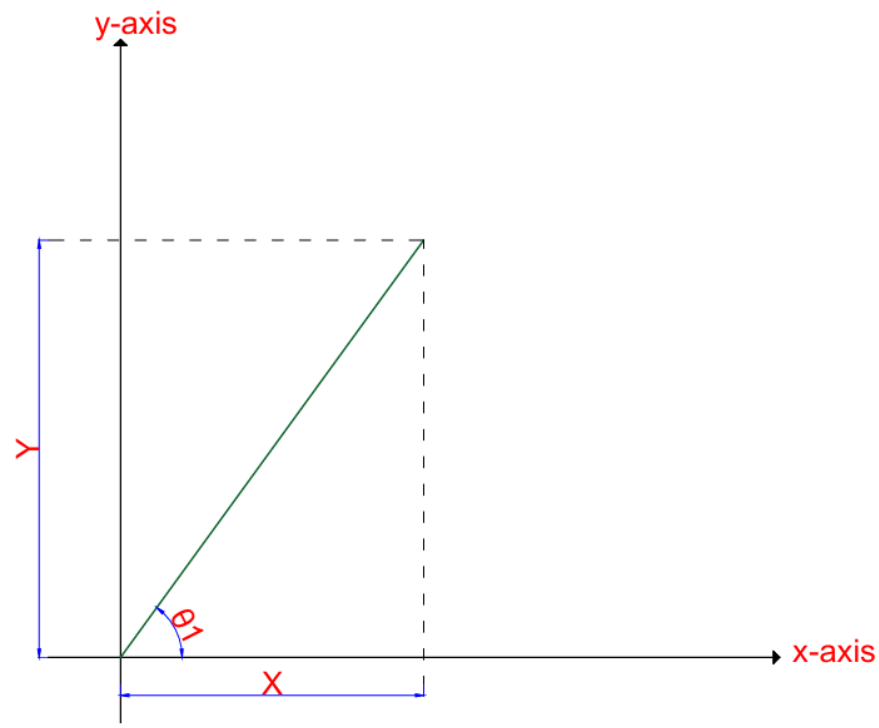


Fig 3: Top View of the Manipulator

3.3 Equation of θ_2

The angle $\boldsymbol{\theta_2}$ is between Link 2 and Link 3. As shown in the figure, a reference configuration is considered in which the base joint angle is set to $\theta_1 = 0°$, and Link-3 (L_3) is parallel to the ground plane. Under this condition, the horizontal position of the end effector is given by,

$$x = L_3 - L_2\cos 45^\circ \text{ ---- (2)}$$

The joint angle $\theta_2$is determined by considering two possible geometric configurations of the manipulator, depending on the position of Link-3 relative to the horizontal axis of Joint-2.

3.3.1 Condition 1: Link-3 below the horizontal axis of Joint-2

When Link-3 is positioned below the horizontal axis of Joint-2 as shown in Fig 4, the vertical position relationship can be expressed as

$$Z + L_4 + b = 55 + L_1 + L_2 \sin\ 45^\circ$$

Rearranging the above equation gives

$$b = 55 + L_1 + L_2 \sin\ 45^\circ - Z - L_4$$

Using the geometric relationship of the triangle formed by Link-3, the joint angle $\theta_2$satisfies

$$\sin\,(45^\circ - \theta_2) = \frac{b}{L_3}$$

Substituting the value of b, the expression becomes

$$\sin\,(45^\circ - \theta_2) = \left(\frac{L_1 + 55 + \frac{L_2}{\sqrt{2}} - Z - L_4}{L_3}\right)$$

Solving for θ_2, the final equation for this configuration is

$$\theta_2 = 45^\circ - arcsin\left(\frac{L_1 + 55 + \frac{L_2}{\sqrt{2}} - Z - L_4}{L_3}\right) \qquad (3)$$

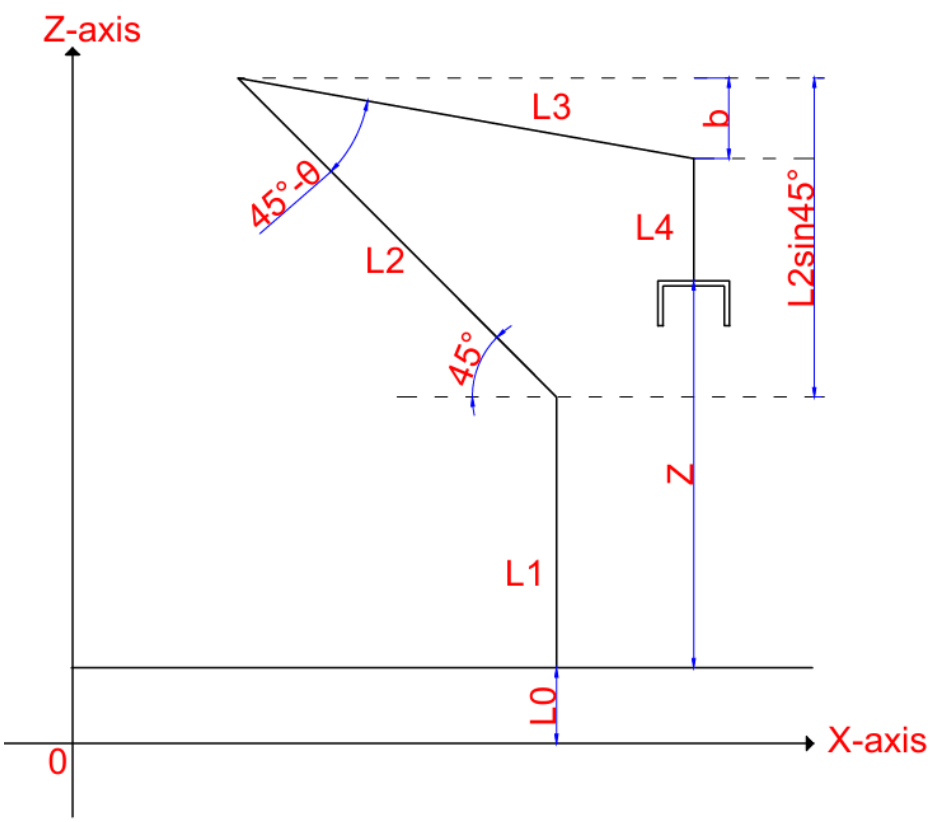


Fig 4: Condition I

3.3.2 Condition 1: Link-3 above the horizontal axis of Joint-2

In the second configuration, Link-3 is located above the horizontal axis of Joint-2 as shown in Fig 5. From the geometric relations shown in the figure, the vertical position equation is given by

$$Z + L_4 = 55 + L_1 + L_2 \sin\ 45^\circ + b$$

Rearranging the equation yields

$$b = L_4 - 55 - \frac{L_2}{\sqrt{2}} + Z - L_1$$

The corresponding trigonometric relationship for the joint angle is

$$\sin\,(\theta_2 - 45^\circ) = \frac{b}{L_3}$$

Substituting the value of b, the expression becomes

$$\sin\,(\theta_2 - 45^\circ) = \left(\frac{L_4 - 55 - \frac{L_2}{\sqrt{2}} + Z - L_1}{L_3}\right)$$

Solving for θ_2, the final equation for this configuration is obtained as

$$\theta_2 = \arcsin\left(\frac{L_4 - 55 - \frac{L_2}{\sqrt{2}} + Z - L_1}{L_3}\right) + 45^\circ \qquad (4)$$

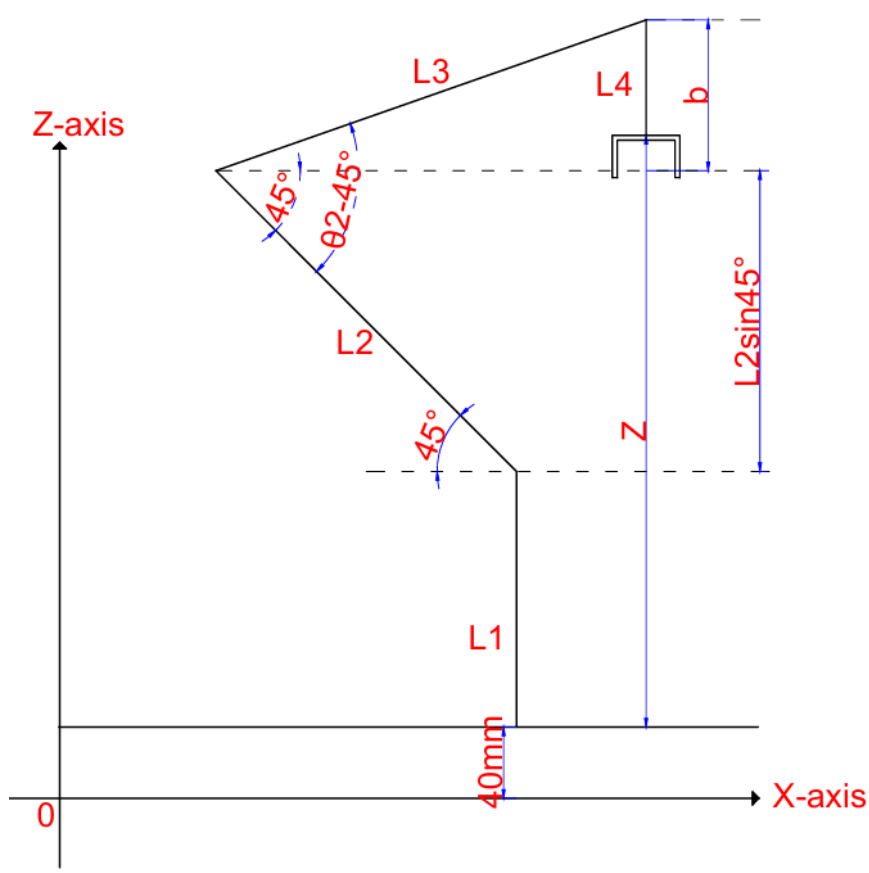


Fig 5: Condition II

3.4 For θ_3

- θ_3 is not part of inverse kinematics because it does not affect the Cartesian position of the end effector; it only performs the gripping operation. It is driven through a gearbox.

4. PID CONTROLLER

A PID (Proportional-Integral-Derivative) controller is a feedback control technique that can be employed to minimize the difference between the desired value and the output. In motor control systems, it is useful in achieving precise positioning.

The standard PID control formula is: $\mathbf{u(t)=K_p\, e(t) + K_i \int e(t)\, dt + K_d\, de(t)/dt}$

where:

- e(t) is the error between the desired and actual position
- K_p is the proportional gain
- K_i is the integral gain
- K_d is the derivative gain
- u(t) is the control input to the motor

Different Variants:

- P-Controller: Output is proportional to the current error. It provides fast response but may cause steady-state errors.
- PI Controller: It eliminates steady-state errors by integrating past errors.
- PD Controller: It improves stability and reduces overshoot.
- PID Controller: It provides all three functions for precise and smooth control.

5. Methodology

1. **Homing procedure:** The base and arm motors were rotated until their respective touch sensors were activated. Once the home position was detected, the motor angles were reset to zero. This ensured a fixed reference position for every cycle and prevented cumulative positioning errors.

2. **Color calibration:** The EV3 color sensor was calibrated by placing each ball near the sensor and recording multiple RGB readings. A tolerance-based method was used to compare measured values with references.

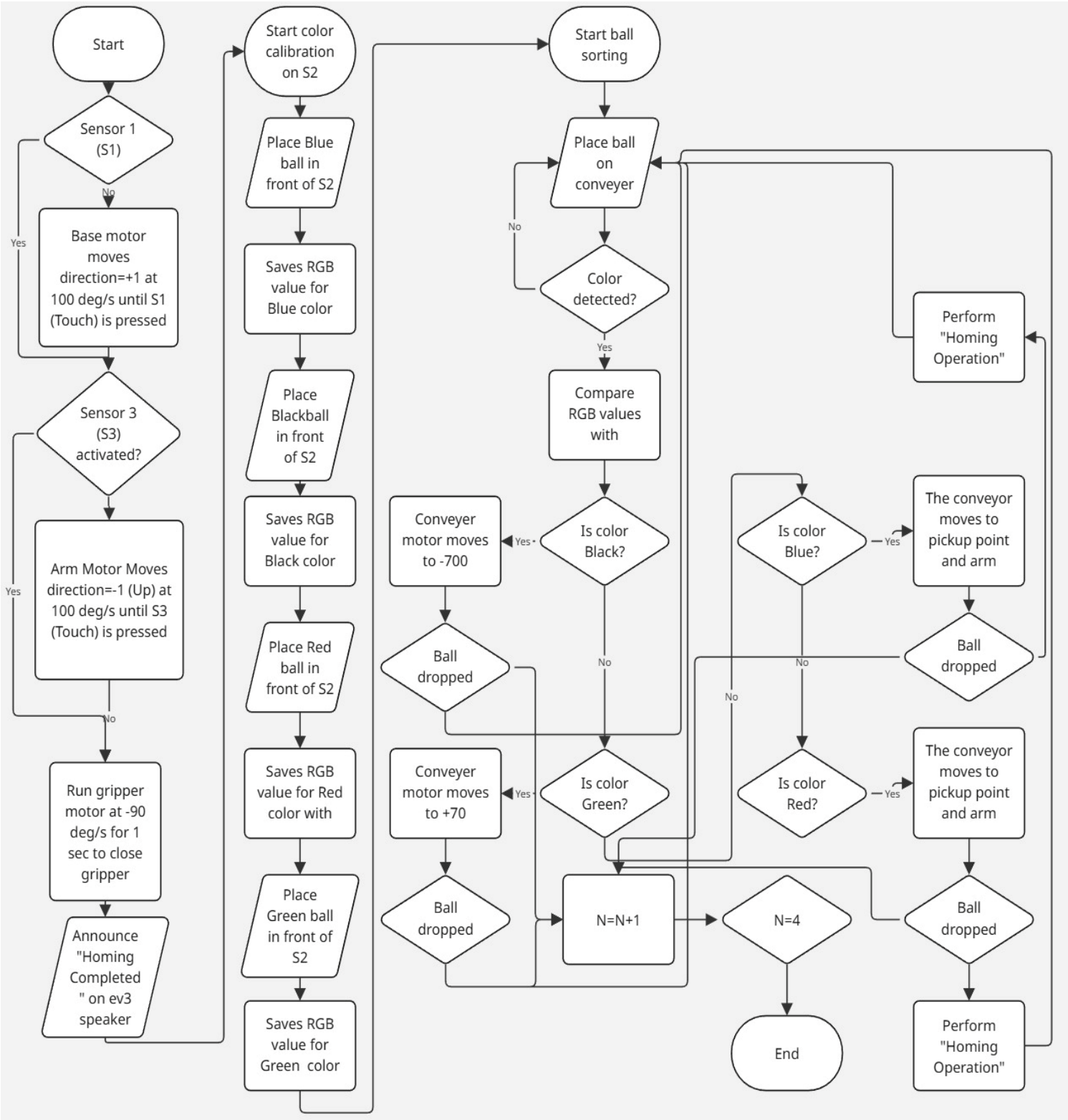
3. **Ball handling:** If the detected color is Black or Green, the conveyor motor rotates to a predefined angle to drop the ball into the corresponding station. If the color is Red or Blue, a full pick-and-place operation is executed.

4. **Inverse kinematics:** Inverse kinematics equations were used to calculate the required arm and base angles for a given target position.

5. **Testing and tuning:** Parameters such as arm coordinates, motor speeds, and tolerance values were adjusted through repeated testing.

6. **Final validation:** The complete system was tested in a continuous loop to verify that all four balls could be sorted correctly and sequentially.

6. Flow Chart


Start
Sensor 1 (S1)
No
Yes
Base motor moves direction=+1 at 100 deg/s until S1 (Touch) is pressed
Sensor 3 (S3) activated?
Yes
Arm Motor Moves direction=-1 (Up) at 100 deg/s until S3 (Touch) is pressed
No
Run gripper motor at -90 deg/s for 1 sec to close gripper
Announce "Homing Completed " on ev3 speaker
Start color calibration on S2
Place Blue ball in front of S2
Saves RGB value for Blue color
Place Blackball in front of S2
Saves RGB value for Black color
Place Red ball in front of S2
Saves RGB value for Red color with
Place Green ball in front of S2
Saves RGB value for Green color
Start ball sorting
Place ball on conveyer
No
Color detected?
Yes
Compare RGB values with
Is color Black?
Yes
Conveyer motor moves to -700
Ball dropped
No
Is color Green?
Yes
Conveyer motor moves to +70
Ball dropped
N=N+1
N=4
End
Is color Blue?
Yes
The conveyor moves to pickup point and arm
Ball dropped
No
Is color Red?
Yes
The conveyor moves to pickup point and arm
Ball dropped
Perform "Homing Operation"
Perform "Homing Operation"